

Yale School of Management - MGT404 Introduction to Microeconomics  
Lectures 11-13 Practice Problems

Page 1-4: questions. Pages 5-7 solutions. **Print Pages 1-4 to work through**, only check solutions afterwards.

1 *Risk Aversion.* A consumer is interested in buying a tablet for \$500, but the tablet has a known failure rate of 5% a year. The consumer has a wealth,  $w$ , of \$10,000 after purchasing the tablet, and is risk averse, with a utility function of  $u(w) = \ln(w)$ . If the tablet fails, the consumer replaces it for \$500. (In an exam setting, we would only use  $u(w) = \sqrt{w}$ ; use a full-feature calculator or set this up in Excel using the function “ln”)

- i. What is the consumer’s certainty equivalent of the current situation?

c: \_\_\_\_\_

- ii. What is the consumer’s risk premium?

*Risk Premium:* \_\_\_\_\_

- iii. What is the maximum amount the consumer would be willing to pay for a warranty that replaces the tablet if the tablet fails in the first year?

*WTP:* \_\_\_\_\_

2 *Wage and Bonus.* A high-end clothing retailer has lately noticed their employees slacking off, and feels sales are suffering as a result. They decide to implement a wage and bonus scheme, to incentivize employees to work hard to make sales. If the employee works hard, the probability of the customer making a purchase is 75%, but if the employee slacks, the probability is only 50%. The retailer knows that its employees can go work at other clothing retailers for a wage of \$18 without needing to put in effort, and that they experience a cost of effort of \$5. Write down the “willing to work” and “do the right thing” constraints with the numbers given above.

- i. *Willing to work:*

\_\_\_\_\_  $\geq$  \_\_\_\_\_

ii. *Do the right thing:*

\_\_\_\_\_  $\geq$  \_\_\_\_\_

iii. What is the optimal wage and bonus for the retailer to set?

*Wage:* \_\_\_\_\_

*Bonus:* \_\_\_\_\_

3. *Drone.* A **risk-neutral** consumer is in the market for a drone to fly around town taking video. The consumer's value for the drone is \$500 if it is operative, and \$0 if it fails. Producer A makes drones with a 5% chance of failure.

(a) (1 points). What is the consumer's willingness to pay for the drone?

*Willingness to Pay:* \_\_\_\_\_

It costs Producer A \$400 to make the drone, and they price it exactly at the consumer's willingness to pay that you found above. Everything is hunky dory until a knockoff drone supplier, Producer B, realizes they can enter the market and make money by selling an inferior product that costs \$200 to make, but that has a 50% failure rate, which is indistinguishable to the consumer from the more reliable drone.

(b) (3 points). If this new entrant were to provide as many drones as the other supplier (i.e. Producer A produces 50% of the market and Producer B produces 50% of the market so that the chance of getting a drone from either company is 50%), what is the consumer's willingness to pay for a random drone they purchase?

*Willingness to Pay:* \_\_\_\_\_

(c) (2 points). Suppose in the short-run, the price is set exactly equal to what you found in part b.

What is the short-term profit per drone of Producer B?

*Producer B's Profit per Drone:* \_\_\_\_\_

What is the short-term profit per drone of producer A?

*Producer A's Profit per Drone:* \_\_\_\_\_

(d) (5 points). Given what you found in (c) and (d), what will be the equilibrium price for drones, and the market share of the two companies?

*Equilibrium Market Price:* \_\_\_\_\_

*Producer A's Market Share:* \_\_\_\_\_

*Producer B's Market Share:* \_\_\_\_\_

(e) (3 points). Producer A can hire an independent testing agency to show that their drones are indeed better than those from Producer B, and only have a 5% failure rate. The consumers will believe the certification of this tester. Assuming there are 1 million potential customers for drones each year, and Producer A would get back 50% of the market with the new certification, how much is Producer A willing to pay for this service annually?

*Max Annual Certification Fee:* \_\_\_\_\_

**Bonus Problem: Production (I received a request for more practice on equating MPIs)**

*Baking up some profits.* Crumbs Bakery, which has been in the news lately, sells cupcakes through retail stores. The production of cupcakes takes both raw ingredients, and labor. The daily production function for cupcakes is given by

$$q = \frac{1}{2} \cdot r \cdot l + l^2 - 20$$

Where output is in dozens of cupcakes,  $r$  is pounds of raw ingredients, and  $l$  is hours of labor. The cost of a pound of raw ingredients is  $p_r = \$1$  while the cost of an hour of labor is  $p_l = \$12$ .

(a) What is the optimal number of pounds of raw ingredients to use per hour of labor?

Pounds of raw ingredients per hour of labor: \_\_\_\_\_

***[NOTE: this problem is from a problem set; the next part would not be asked on an exam. You can compute the solution via Excel Solver if interested]***

(b) A standard day requires about 25 dozen cupcakes to be made. What is the cost-minimizing production plan for 25 dozen cupcakes? How much will this cost?

Pounds of raw ingredients: \_\_\_\_\_

Hours of labor: \_\_\_\_\_

Total cost: \_\_\_\_\_

## SOLUTIONS

**Advice: ALWAYS attempt problems on your own, then discuss with learning team, before looking at solutions.**

### Problem 1:

(i) c: 9974.3862

(ii) Risk Premium: 0.6138

(iii) WTP: 25.6138

$$\begin{aligned} &= \\ \text{EU} &= 0.05 \times \log(10000 - 500) + 0.95 \times \log(10000) & (1) \\ c &= \exp(\text{EU}) = 9974.3862 & (2) \end{aligned}$$

$$\text{EV} = 0.05 \times (10000 - 500) + 0.95 \times (10000) = 9975 \quad (3)$$

$$\text{RiskPremium} = \text{EV} - c = 0.6138 \quad (4)$$

$$\text{WTP} = 10000 - c = 25.6138 \quad (5)$$

To check: note that if the price for insurance were \$25.6138, and I buy it, I am guaranteed utility  $\ln(10000 - 25.6138)$ , which is the same as the expected utility of the risk; therefore I would just be willing to pay that.

### Problem 2:

(i) Sign the contract:  $w - 5 + 0.75b \geq 18$

(ii) Do the right thing:  $w - 5 + 0.75b \geq w + 0.5b$

(iii) Wage: 8, Bonus: 20

First, set the wage to just satisfy the first constraint

$$\begin{aligned} w &= 18 + 5 - 0.75b \\ &= 23 - 0.75b \end{aligned}$$

Then, solve for the bonus in the 2<sup>nd</sup> constraints

$$\begin{aligned} w - 5 + 0.75b &= w + 0.5b \\ 0.25b &= 5 \\ b &= 20 \end{aligned}$$

Subbing back into wage, we get  $w = 23 - 0.75(20) = 8$

**Problem 3:**

(a) (1 pt)

WTP: 475

$$WTP_A = 0.05 \times 0 + 0.95 \times 500 = 475$$

(b) (3 pt)

WTP: 362.5

WTP for product B is:

$$WTP_B = 0.5 \times 0 + 0.5 \times 500 = 250$$

WTP for a randomly selected drone is:

$$WTP = 0.5 \times WTP_A + 0.5 \times WTP_B = 0.5 \times 475 + 0.5 \times 250 = 362.5$$

(c) (2 pt)

Producer B's Profit per Drone: 162.5

Producer A's Profit per Drone: -37.5

B makes  $362.5 - 200 = 162$  per drone while A makes  $362.5 - 400 = -37.5$  per drone.

(d) (5 pt)

Equilibrium Market Price: 250

Producer A's Market Share: 0%

Producer B's Market Share: 100%

A will exit the market and only B will remain. The equilibrium price will be equal to a consumer's WTP for B, which is  $WTP_B = 250$ .

(e) (3 pt)

Max Annual Certification Fee: 37,500,000

With the certification, A can set price equal to a consumer's WTP for A, which is  $WTP_A = 475$ .

Since A is assumed to take 0.5M consumers, its profit will be  $(475 - 400) \times 0.5M = 37.5M$ .

A makes zero profit without the certification, so it will pay up to 37.5M for the process.

### Bonus Problem (Production) Answer

The marginal products of raw ingredient and labor are given by

$$MP_r = \frac{\partial q}{\partial r} = \frac{1}{2}l$$

$$MP_l = \frac{\partial q}{\partial l} = \frac{1}{2}r + 2l$$

with marginal productivity of investment

$$MPL_r = \frac{MP_r}{p_r} = \frac{\frac{l}{2}}{1} = \frac{l}{2}$$

$$MPL_l = \frac{MP_l}{p_l} = \frac{\frac{1}{2}r + 2l}{12} = \frac{r + 4l}{24}$$

To solve for the optimal mix of inputs, we equate the marginal productivities of investment for all our inputs:

$$MPL_r = MPL_l$$

$$\frac{l}{2} = \frac{r + 4l}{24}$$

$$12l = r + 4l$$

$$r = 8l$$

So Crumbs should use 8 pounds of raw ingredients for every hour of labor.

The excel solution to part (b) is:

Pounds of raw ingredients: 24\_\_\_\_\_

Hours of labor: 3\_\_\_\_\_

Total cost: \$60\_\_\_\_\_